

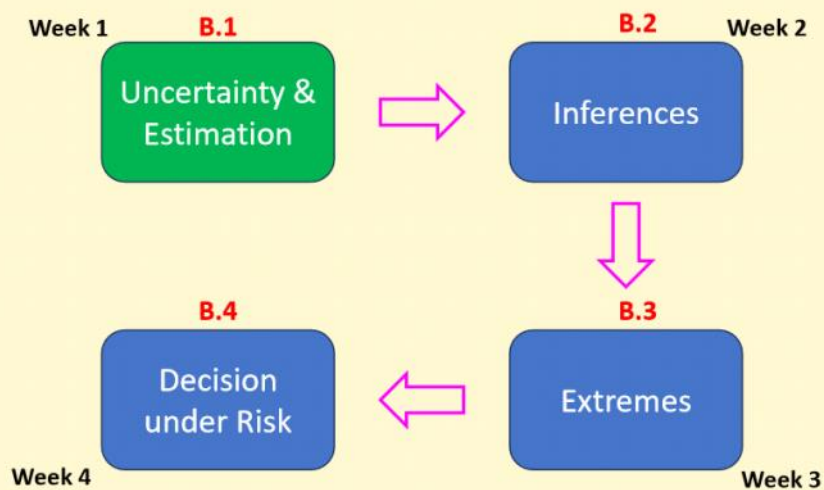
CV 5101

Modeling, Uncertainty, and Data for Engineers

(July – Nov 2026)

Dr. Prakash S Badal

Module Overview



A crash course in probability, probabilistic models, and probabilistic methods.

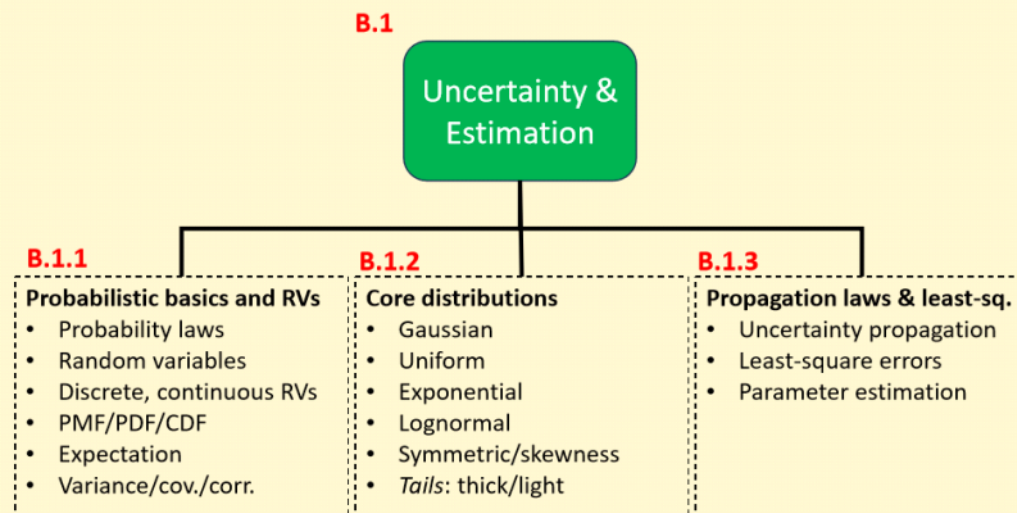
Check-in with teach-book

<https://mude.citg.tudelft.nl/book/2024/>

- Q1 Topics (Chapters 5, 6, 2, and 3)
 - **Q1C5 Univariate continuous distribution**
 - Q1C5.1 PDF/CDF
 - Q1C5.2 Empirical Distributions
 - Q1C5.3 Parametric Distributions
 - Q1C5.4 Fitting a Distribution
 - Q1C6 Multivariate Distributions (briefly)
 - **Q1C2 Propagation of Uncertainty**
 - Q1C3 Observation Theory: least-sq., Hyp. Test, Conf. Intervals
- Q2 Topics (Chapter 7 and 8)
 - Q2C7 Extreme value theory: GEV, return period, POT
 - Q2C8 Risk and decision making (CBA)
- Fundamental Concepts
 - **Chapter 6, 7, 8, 9. Probability basics, rv, z- and t-tables**
- Programming
 - Fundamental Concepts → Chapter 10

3

Module Overview



B.1.1 Probability basics & random variables

Birthdays!

Birthday-Birthday

Jan 4 14	Feb 22 23	Mar 12 14
Apr 6 15 17	May 6 11 12 14 23	Jun 25 26
Jul	Aug	Sep
Oct	Nov	Dec

7

Question 1

Probability that
at least two
people in this
room share the
birthday?



slido.com
#cv5101

Birthday Problem

$$\Pr(\text{Two people share birthday}) = 1/365$$

$$\Pr(2 \text{ people share birthday in a class of } 40)$$

$$= \frac{364 \times 363 \times \dots \times (365 - n + 1)}{365 \times 365 \times \dots \text{ } 40 \text{ times}}$$

for $n=50$, $\Pr(\text{at least 2 shared birthdays}) \approx 0.97$
 $n=23$, $\Pr(\text{at least 2 shared birthdays}) \approx 0.5$

Birthday attack $\leftarrow$ collision attack

$$2^{64} \longrightarrow 2^{32}$$

Intuition can be misleading.

9

Settlers of Catan



[Wikimedia commons](#)



[colonist.io](#) with permission

10

Practice question

- Work with the **another person around you**.
- You throw 2 dice;
- add the values of each die
- List all the possible outcomes



Q1. What is the probability that **the sum of the two dice will be 6?**

Q2. What is the probability that you get **at least one 8 in a single round**

consisting of four players? $1 - \left(\frac{31}{36}\right)^4$

slido.com
#cv5101

Probability axioms

S : Sample space

$$\Pr(S) = 1$$

E : An event

$$0 \leq \Pr(E) \leq 1$$

mutually
exclusive

E_1, E_2 : Two events s.t. $E_1 \cap E_2 = \emptyset$

$$\Pr(E_1 \cup E_2) = \Pr(E_1) + \Pr(E_2)$$

Probability rules

$$Pr(E_1 \cup E_2) = Pr(E_1) + Pr(E_2) - Pr(E_1 \cap E_2)$$

Union ↑

$$Pr(\bar{E}_1) = 1 - Pr(E_1)$$

Complement ↑

$$Pr(E_1 \cup E_2 \cup E_3) = Pr(E_1) + Pr(E_2) + Pr(E_3) - Pr(E_1 \cap E_2) - Pr(E_2 \cap E_3) - Pr(E_3 \cap E_1) + Pr(E_1 \cap E_2 \cap E_3)$$

Inclusion - Exclusion principles

$$\equiv Pr(E_i \cap E_j)$$

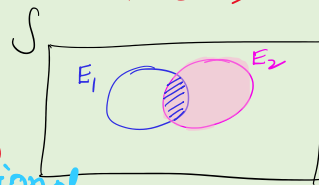
$$Pr\left(\bigcup_{i=1}^n E_i\right) = \sum Pr(E_i) - \sum_{i < j} Pr(E_i \cap E_j) + \sum_{i < j < k} Pr(E_i \cap E_j \cap E_k) - \dots + (-1)^{n-1} Pr(E_1 \cap E_2 \cap \dots \cap E_n)$$

13

Probability rules: conditional prob.

- Conditional probability: $Pr(E_1 | E_2) = \frac{Pr(E_1 \cap E_2)}{Pr(E_2)}$

Probability of E_1 given that E_2 has occurred



- Multiplication rule:

$$Pr(E_1 \cap E_2) = Pr(E_1 | E_2) \cdot Pr(E_2)$$

$$Pr(E_1 \cap E_2 \cap \dots \cap E_n) = Pr(E_1 | E_2 E_3 \dots E_n) \cdot Pr(E_2 | E_3 E_4 \dots E_n) \cdot \dots \cdot Pr(E_{n-1} | E_n) \cdot Pr(E_n)$$

Margin ↓

- Two events are called statistically independent (SI), if

$$Pr(E_1 | E_2) = Pr(E_1)$$

$$\Rightarrow \frac{Pr(E_1 \cap E_2)}{Pr(E_2)} = Pr(E_1) \Rightarrow Pr(E_1 \cap E_2) = Pr(E_1) \cdot Pr(E_2)$$

↑ Non-zero unless either $Pr(E_1)$, $Pr(E_2)$ is zero

L2

Sep 2, 2026

Messages from the Last Class

- **Message 1: Space Shuttle Challenger Disaster**

If everything is important, then nothing is!

- **Message 2: Basic Probability**

→ you need to revise prob & stats

→ Sample space, S & Event, E

$$Pr(S) = 1 \quad 0 \leq Pr(E) \leq 1$$

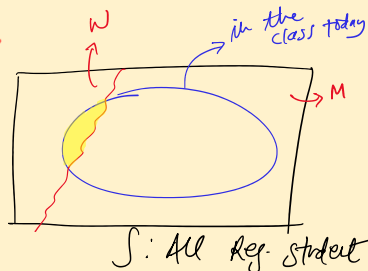
If E_1 & E_2 are exclusive $E_1 \cap E_2 = \emptyset$

$$Pr(E_1 \cup E_2) = Pr(E_1) + Pr(E_2)$$

- **Message 3: Conditional prob.**

$$\rightarrow Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}$$

→ Conditioning is equivalent to reducing the sample space.



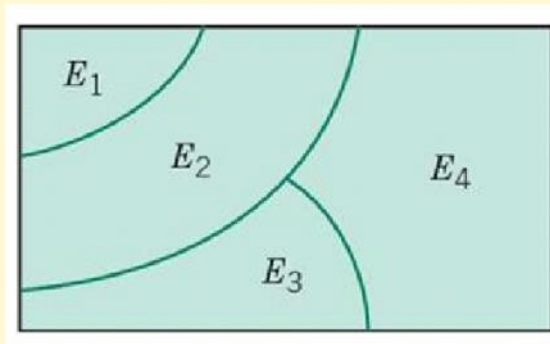
Set Theory: Mutually Exclusive Events

If $E_1, E_2, \dots, E_n$ are such that

$$E_i \cap E_j = \emptyset$$

then

$$Pr(E_1 \cup E_2 \cup \dots \cup E_n) = Pr(E_1) + Pr(E_2) + \dots + Pr(E_n)$$



15

Probability rules: total probability rule

© ADK22

- If $E_1, E_2, \dots, E_n$ are n MECE events, Mutually exclusive & collectively exhaustive
 $E_1 \cup E_2 \cup \dots \cup E_n = S$



$$\begin{aligned} Pr(A) &= Pr(AE_1) + Pr(AE_2) + \dots + Pr(AE_n) \\ &= \sum_{i=1}^n Pr(AE_i) \\ &= \sum_{i=1}^n Pr(A|E_i) \cdot Pr(E_i) \end{aligned}$$

↑ Total Probability Rule

16

(At home) Probability rules: total probability rule

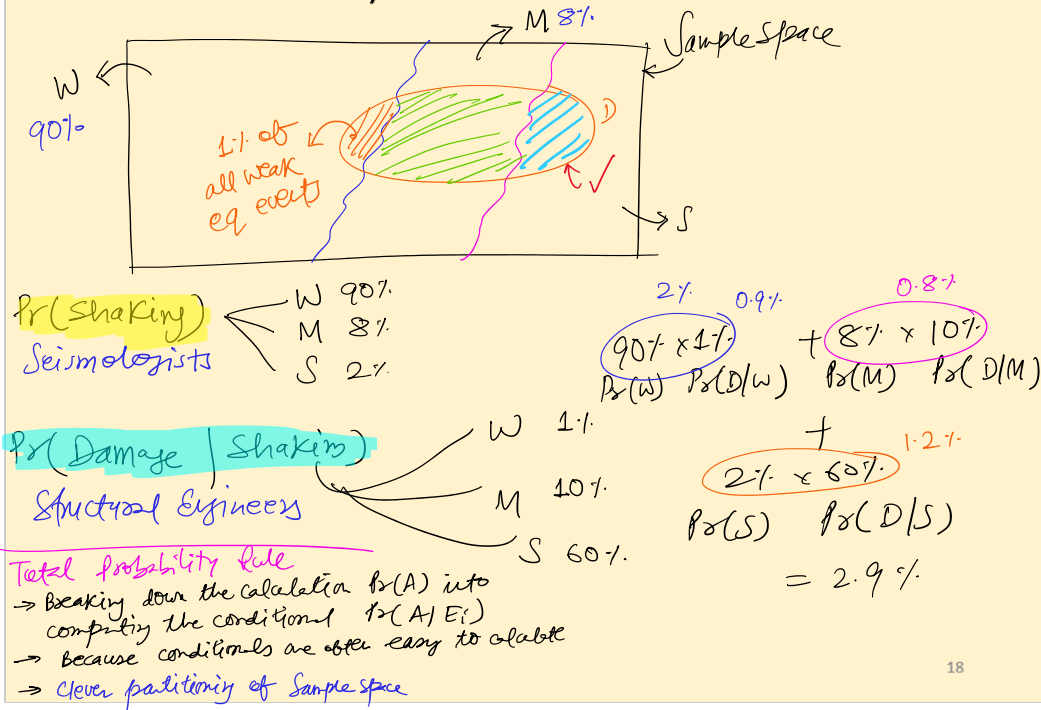
The probability that a building will sustain damage due to an earthquake depends on the intensity of shaking. Suppose the intensity is discretized into the states of weak, moderate, and strong. Assume it is determined that the probability of damage to the building is 0.01, 0.10, and 0.60 for the three levels of shaking intensity, respectively. Furthermore, based on the history of past earthquakes in the region, it is estimated that, for a randomly occurring earthquake, the probabilities for the three levels of shaking intensity at the site of the building are 0.90, 0.08, and 0.02, respectively. We are interested in the probability that the building will sustain damage during the next earthquake in the region.

© ADK22

$$Pr(D|W)$$

17

Total Probability Rule: Behind the Scenes!



18

Partitioning Complex Structural Systems

probability that Chennai
Airport is closed tomorrow?

Weather prediction
 → No rain
 → Moderate rain
 → Heavy rain
 → Cyclone

Prob. that a high-rise building
cannot be evacuated safely
during a fire?

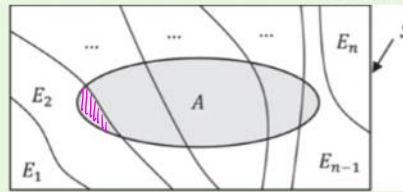
Fire initiation
 → Basement
 → Lower floor
 → Middle floor
 → Upper floor

19

Probability rules: Bayes' rule

• If $E_1, E_2, \dots, E_n$ are n MECE events,

$P(E_i|A)$ ← Probability of Interest



© ADK22

$$P(A \cap E_i) = P(E_i|A) \cdot P(A) = P(A|E_i) \cdot P(E_i)$$

$$P(E_i|A) = \frac{P(A|E_i) \cdot P(E_i)}{P(A)}$$

$$= \frac{P(A|E_i) \cdot P(E_i)}{\sum_{j=1}^n P(A|E_j) \cdot P(E_j)}$$

20

(At home) Probability rules: Bayes' rule

- Assume in the previous example that the building has sustained damage. What can we learn about the intensity of the earthquake?

$$\begin{array}{lcl}
 P_r(W|D) = 0.9/2.9 = 31\% & | & 90\% \leftarrow P_r(W) \\
 P_r(M|D) = 0.8/2.9 = 27\% & | & 8\% \leftarrow P_r(M) \\
 P_r(S|D) = 1.2/2.9 = 41\% & | & 2\% \leftarrow P_r(S)
 \end{array}$$

$\uparrow$ posterior $\uparrow$ prior

$$\begin{array}{c}
 P_r(W|D) \\
 P_r(M|D) \\
 P_r(S|D)
 \end{array}
 \quad \vdots \quad
 \begin{array}{c}
 P_r(W) \\
 P_r(M) \\
 P_r(S)
 \end{array}$$

$\nwarrow$

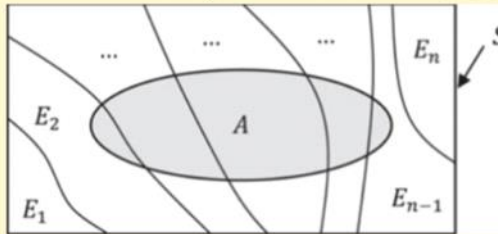
Posterior = likelihood $\times$ prior

21

Total Probability Rules & Bayes' rule

© ADK22

$E_1, E_2, \dots, E_n$ are (MECE) partitions of S .



22

L3

Sep 3, 2026

Total Probability and Bayes' Rule (irl)

Total Probability Rule Q) What will happen?

- $\Pr(\text{Cancer}) \rightarrow \text{Insurance}$
- $\Pr(\text{flight cancelled}) \rightarrow \text{Flight operator}$
- $\Pr(\text{Building collapse}) \rightarrow \text{Design standard}$

Bayes' Rule Q) ^{Now that "this" has happened,} What caused this event?

- Given that a patient has some symptom
 - Request more tests
 - Treatment/Medication depends on these prob.

Murder trial of O. J. Simpson

O. J. Simpson

57 languages

Article Talk

Read

Edit source

View history

☆ Watch

From Wikipedia, the free encyclopedia

"The Juice" redirects here. For other uses, see Juice (disambiguation).

Orenthal James Simpson (July 9, 1947 – April 10, 2024), nicknamed "**the Juice**",^[b] was an American **football** player, actor, and media personality. He played in the **National Football League** (NFL) for 11 seasons—nine with the **Buffalo Bills**—and is regarded as one of the greatest **running backs** of all time. His success is widely considered to be overshadowed by his two criminal charges for the **1994 murders** of his ex-wife **Nicole Brown** and her friend **Ron Goldman**, and the **contentious criminal trial** in which he was acquitted on both counts.

Simpson played **college football** for the **USC Trojans**, winning the 1968 **Heisman Trophy** as a **senior**. He was selected **first overall** by the Bills in the **1969 NFL/AFL draft**. With the Bills, he received five consecutive **Pro Bowl** and first-team **All-Pro** selections from 1972 to 1976. He led the league in **rushing yards** four times, in rushing **touchdowns** twice, and in points scored in 1975. Simpson became the first NFL player to rush for more than **2,000 yards** in a season—earning him 1973's **NFL Most Valuable Player (MVP)** award—and he is the only player to do so in a 14-game **regular season**. He also holds the record for the single-season yards-per-game average, at 143.1.

O. J. Simpson



Murder trial of O. J. Simpson

Which probability should the jury care about?

$\Pr(\text{DNA match} \mid \text{innocent})$

Or

$\Pr(\text{guilty} \mid \text{DNA match})$



15-20x
more

Murder trial of O. J. Simpson

DNA expert say, that

$$\Pr(\text{DNA match} | \text{unrelated person}) = \frac{1}{20 \text{ million}}$$

Jury wants to know

$$\Pr(\text{guilty} | \text{DNA match})$$

Bayes' $\Pr(\text{guilty} | \text{DNA match}) = \frac{\Pr(\text{DNA match} | \text{guilty}) \cdot \Pr(\text{guilty})}{\Pr(\text{DNA match})}$

Prosecutor's Bias $\Pr(G | \text{DNA}) \propto \Pr(\text{DNA} | G)$ ← NOT wrong, But incomplete

$$\Pr(G | \text{DNA}) \propto \Pr(\text{DNA} | G) \cdot \underbrace{\Pr(G)}_{\text{Baseline rate}}$$

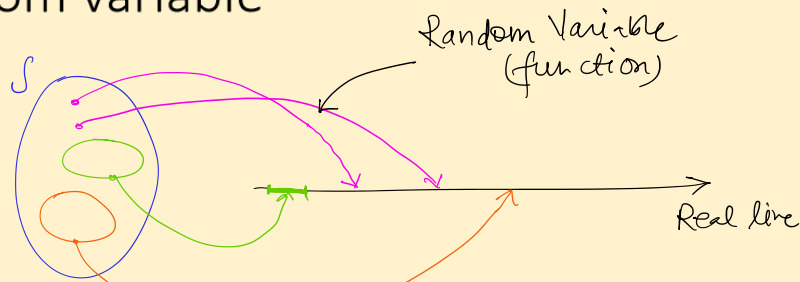
DNA match $\Rightarrow$ Create guilt

100,000,000

	MATCH	NO MATCH
GUILTY	1	0
INNOCENT	5	~100,000,000

Random Variables

Random variable



Experiment: Tossing two coins

Sample space $\{(H,T), (H,H), (T,H), (T,T)\}$

X : Number of heads in two tosses of a coin

$$X(\omega) = \begin{cases} 0 & ; \quad \omega = (T, T) \\ 1 & ; \quad \omega = (H, T) \text{ or } (T, H) \\ 2 & ; \quad \omega = (H, H) \end{cases}$$

↑ Random Variable

RV: misnomer. Neither ^{Random} Variable. It is a function.

28

Random variable

- Experiment: rolling two 4-faced dice

X = maximum roll

Domain of X ? $\rightarrow S$

Range of X ? $\rightarrow \{1, 2, 3, 4\}$

$S = ?$

$X(\omega) = ?$

$$X(\omega) = \begin{cases} 1, & \omega = (1, 1) \\ 2, & \omega = (1, 2), (2, 1), (2, 2) \\ 3, & \omega = (1, 3), (2, 3), (3, 1), (3, 2), (3, 3) \\ 4, & \omega = (1, 4), (2, 4), (3, 4), (4, 1), (4, 2), (4, 3), (4, 4) \end{cases}$$

$$P_X(X=1) = 1/16$$

$$P_X(X=2) = 3/16$$

$$P_X(X=3) = 5/16$$

$$P_X(X=4) = 7/16$$

$$\sum_{i=1}^n P_X(X=\omega_i) = 1$$

29

Types of RV

Discrete

- Range is finite or countable
 - Grade in this course
 - # of sand particles on Marina
 - Natural numbers

Continuous

- Range is continuous over some interval
 - temperature
 - height
 - weight
 - $x \in (0, 4)$

30

L4
Sep 8, 2026

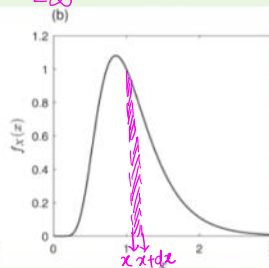
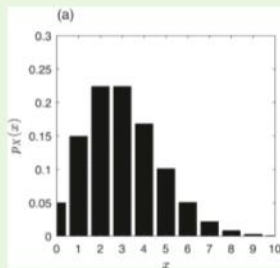
Random variable (rv): PMF, PDF

- For a discrete rv:

Probability mass function (PMF) $P_X(x) = \Pr(X=x)$
 PMF must satisfy: $0 \leq P_X(x) \leq 1$, $\sum_x P_X(x) = 1$

- For a continuous rv:

Probability density function (PDF) $f_X(x)$
 $\Pr(x < X \leq x+dx) = \int_x^{x+dx} f_X(x) dx$
 PDF must satisfy: $f_X(x) \geq 0$, $\int_{-\infty}^{\infty} f_X(x) dx = 1$



© ADK22

31

SIDE NOTE:

for real numbers,
 Never compare their equality

~~if $x = 5$~~

if $\text{abs}(x-5) < \text{tol}$ ✓

(At home) Random variable (rv): PMF, PDF

- Let X = concrete strength, in the range $[0, 20 \text{ MPa}]$

- Assume a PDF given as:

$$f_X(x) = 0.05 \text{ for } 0 \leq x \leq 20$$

- What is the probability that the strength of a tested concrete is less than 10 MPa?

32

Random variable (rv): CDF

- Cumulative distribution function (CDF) is defined as:

$$F_X(x) = \Pr(X \leq x)$$

- The CDF of a **discrete RV** X is

$$F_X(x) = \sum_{x_i \leq x} p_X(x_i) \rightarrow \Pr(X \leq x)$$

- The CDF of a **continuous RV** X is

$$F_X(x) = \int_{-\infty}^x f_X(x) dx \rightarrow \begin{aligned} &f_X(x) = \frac{d}{dx} F_X(x) \\ &\text{Non-decreasing} \end{aligned}$$

$$\text{CCDF} = \Pr(X > x) = 1 - F_X(x)$$

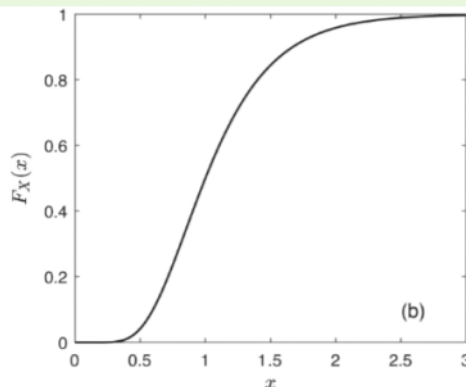
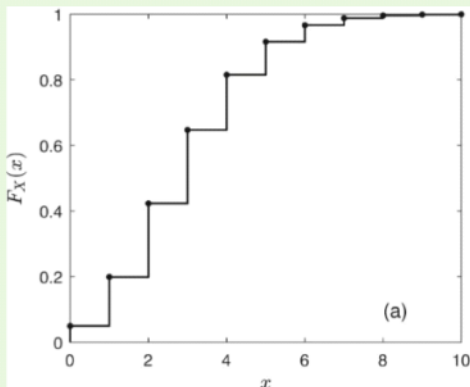
↳ Complementary

33

Random variable (rv): CDF

CDF must satisfy: $F_X(-\infty) = 0$; $F_X(\infty) = 1$

$$\text{if } x < y \Rightarrow F_X(x) \leq F_X(y)$$



© ADK22 34

	fair	loaded/biased
1	1/6	$\alpha = 1/12$
2	1/6	$3\alpha = 3/12$
3	...	α ...
4	...	3α ...
5	...	α ...
6	1/6	3α ...
		$12\alpha = 1$
		$\alpha = 1/12$

Expectation of RV

Expectation

$$E[X] = \sum x_i p_X(x_i) = \int_{-\infty}^{\infty} x f_X(x) dx$$

- Let $g(X)$: a function of rv X , with PDF $f_X(x)$.

Expectation of $g(X)$,

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

$$E[\sum_i g_i(X)] = \sum_i E[g_i(X)]$$

→ expectation is a linear operator

- Expectation/mean of X

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

measure of centrality

→ deviation $X - \mu \rightarrow E[X - \mu] = 0$

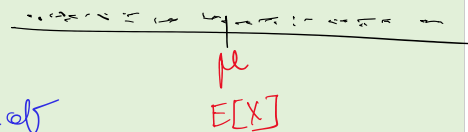
→ Mean Absolute Deviation (MAD)

$$E[|X - \mu|]$$

→ Square of deviation →

$$E[(X - \mu)^2] = \sigma^2$$

↓
Variance



Variance

- Variance is the expected value of $(X - \mu_X)^2$,

$$\begin{aligned}\sigma^2 &= E[(X - \mu)^2] \\ &= E[X^2 + \mu^2 - 2X\mu] \\ &= E[X^2] + E[\mu^2] - E[2X\mu] \\ &= E[X^2] + \mu^2 - 2\mu^2 \\ &= E[X^2] - \mu^2 \\ &= E[X^2] - (E[X])^2\end{aligned}$$

37

Covariance

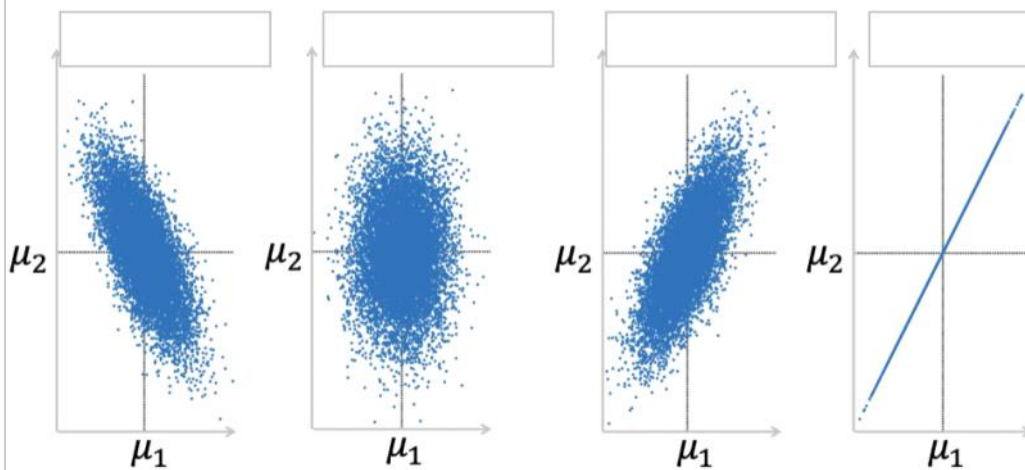
38

Correlation

- Correlation coefficient

39

Correlation



40

Analogy with Engineering Mechanics

Param.	Mechanics	Probability

Questions, comments,
or concerns?